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Subject: ASD & DevOps

EXPERIMENT:-3

Aim: To deploy multi-service applications using docker compose

This Experiment maps to **LO4**

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory:

Deployment of Multi-Service Applications Using Docker Compose

Docker Compose is a utility that simplifies the deployment and management of applications that require multiple Docker containers. Instead of creating and running each container separately, Docker Compose allows all services to be defined in a single YAML configuration file (compose.yaml or docker-compose.yml) and managed together using simple commands.

A multi-service application is made up of several independent services that work together to provide complete functionality. Common examples include a frontend, backend, database, and cache. Each service runs inside its own Docker container, ensuring isolation, portability, and consistency across different systems.

In this experiment, Docker Compose is used to deploy an application with four services:

Frontend Service (Nginx)

- Provides the user interface of the application.
- Hosts and serves static web pages.
- Sends requests to the backend whenever application data is required.

Backend Service (Python Flask)

- Implements the application's business logic.
- Processes requests received from the frontend.
- Communicates with the MySQL database and Redis cache.

Database Service (MySQL)

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- Stores and manages the application's data.
- Uses Docker volumes to ensure that data remains available even if the container is stopped or recreated.

Redis Service

- Functions as an in-memory caching system.
- Stores frequently accessed data to improve application speed and reduce database load.

Docker Compose uses a YAML file to specify:

- Application services and their configurations
- Docker images or build instructions
- Port mappings
- Environment variables
- Service dependencies
- Persistent storage volumes

Some commonly used Docker Compose commands include:

Command	Description
mkdir myapp	Creates a new project directory named myapp to store the application files.
cd myapp	Moves into the project directory where Docker Compose files and service folders are created.
mkdir frontend	Creates a directory for the frontend service.
mkdir backend	Creates a directory for the backend service.
notepad docker-compose.yml / notepad compose.yaml	Creates and opens the Docker Compose configuration file where all services are defined.
dir	Lists files and folders in the current directory to verify that required files exist.

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docker --version	Checks the installed Docker version.
docker compose version	Checks the Docker Compose version installed on the system.
docker compose config	Validates the Compose YAML file and displays the final configuration Docker will use.
docker compose build	Builds Docker images for services that use custom Dockerfiles. In this project, it built the frontend and backend images.
docker compose up	Creates and starts all containers defined in the Compose file and displays live logs.
docker compose up -d	Starts all services in detached mode (runs containers in the background).
docker compose ps	Displays the status of all services managed by Docker Compose.
docker ps	Shows currently running Docker containers.
docker ps -a	Shows all containers, including stopped and created containers.
docker images	Lists Docker images available on the system.
docker compose images	Displays images used by the current Compose project.
docker compose logs	Displays logs from all running services for monitoring and troubleshooting.
docker compose logs frontend	Displays logs generated by the frontend service.

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docker compose logs backend	Displays logs generated by the backend service.
docker compose logs database	Displays MySQL database service logs.
docker compose logs redis	Displays Redis service logs.
docker compose stop	Stops running containers without removing them.
docker compose start	Starts previously stopped containers.
docker compose restart	Restarts all services.
docker compose down	Stops and removes containers, networks created by Docker Compose.
docker compose down -v	Stops and removes containers, networks, and associated volumes (also removes stored database data).
docker logs <container_name>	Displays logs of a specific container.
docker start <container_name>	Starts an existing stopped container manually.
docker pull mysql:8.0	Downloads the MySQL 8.0 image from Docker Hub.
docker pull redis:latest	Downloads the latest Redis image from Docker Hub.

OUTPUT:

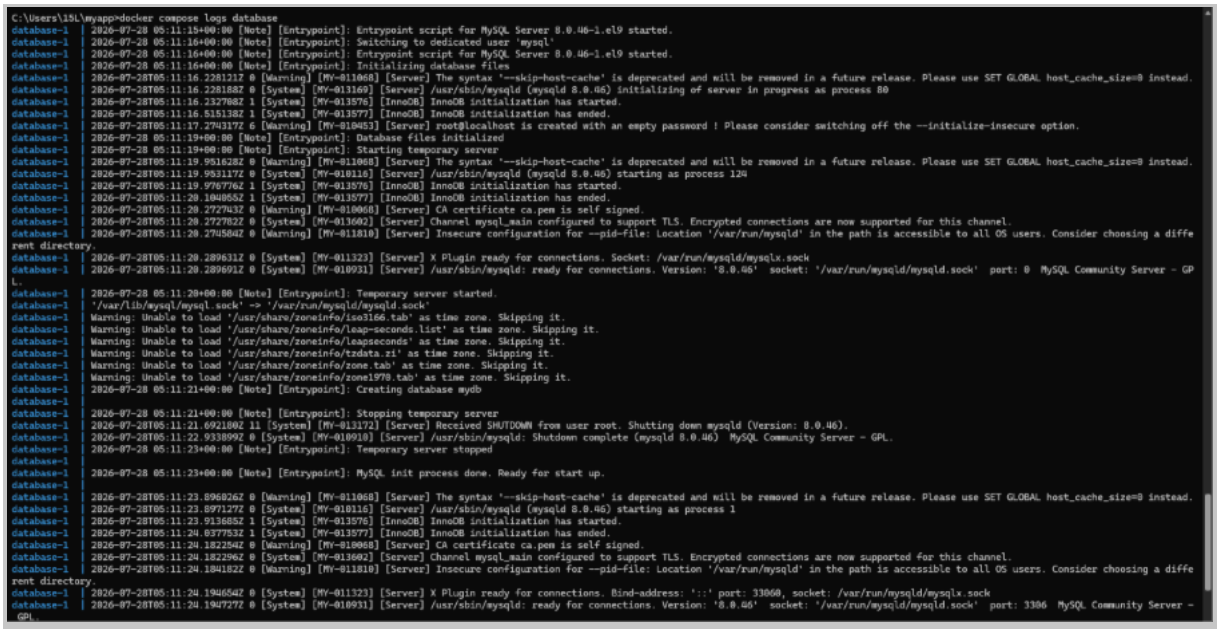


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```
C:\Users\15L\myapp>docker compose logs frontend
frontend-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform confi
guration
frontend-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default
.conf
frontend-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/defau
lt.conf
frontend-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: using the "epoll" event method
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: nginx/1.31.3
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker processes
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 29
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 30
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 31
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 32
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 33
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 34
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 35
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 36
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 37
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 38
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 39
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 40
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 41
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 42
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 43
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 44
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 45
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 46
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 47
frontend-1  | 2026/07/28 05:11:16 [notice] 1#1: start worker process 48
frontend-1  | 172.18.0.1 - - [28/Jul/2026:05:14:35 +0000] "GET / HTTP/1.1" 200 137 "-" "Mozilla/5.0 (W
indows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36" "-"
frontend-1  | 172.18.0.1 - - [28/Jul/2026:05:14:35 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://
localhost/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/1
50.0.0.0 Safari/537.36" "-"
frontend-1  | 2026/07/28 05:14:35 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed
(2: No such file or directory), client: 172.18.0.1, server: localhost, request: "GET /favicon.ico HTTP
/1.1", host: "localhost", referrer: "http://localhost/"
```

```
C:\Users\15L\myapp>docker compose logs backend
backend-1  |  * Serving Flask app 'app'
backend-1  |  * Debug mode: off
backend-1  | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
backend-1  |  * Running on all addresses (0.0.0.0)
backend-1  |  * Running on http://127.0.0.1:5000
backend-1  |  * Running on http://172.18.0.4:5000
backend-1  | Press CTRL+C to quit
backend-1  | 172.18.0.1 - - [28/Jul/2026 05:14:52] "GET / HTTP/1.1" 200 -
backend-1  | 172.18.0.1 - - [28/Jul/2026 05:14:52] "GET /favicon.ico HTTP/1.1" 404 -
```

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```
C:\Users\15L\myapp>docker compose logs database
database-1  | 2026-07-28 05:11:15+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
database-1  | 2026-07-28 05:11:16+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
database-1  | 2026-07-28 05:11:16+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
database-1  | 2026-07-28 05:11:16+00:00 [Note] [Entrypoint]: Initializing database files
database-1  | 2026-07-28T05:11:16.228121Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1  | 2026-07-28T05:11:16.228188Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.46) initializing of server in progress as process 80
database-1  | 2026-07-28T05:11:16.232788Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:11:16.515138Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1  | 2026-07-28T05:11:17.274317Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
database-1  | 2026-07-28 05:11:19+00:00 [Note] [Entrypoint]: Database files initialized
database-1  | 2026-07-28 05:11:19+00:00 [Note] [Entrypoint]: Starting temporary server
database-1  | 2026-07-28T05:11:19.951628Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1  | 2026-07-28T05:11:19.953117Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 124
database-1  | 2026-07-28T05:11:19.976776Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:11:20.104865Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1  | 2026-07-28T05:11:20.272743Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1  | 2026-07-28T05:11:20.272782Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1  | 2026-07-28T05:11:20.275584Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a diffe
rent directory.
database-1  | 2026-07-28T05:11:20.289631Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
database-1  | 2026-07-28T05:11:20.289691Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 0  MySQL Community Server - GP
L.
database-1  | 2026-07-28 05:11:20+00:00 [Note] [Entrypoint]: Temporary server started.
database-1  | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
database-1  | Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
database-1  | Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
database-1  | Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
database-1  | Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
database-1  | Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
database-1  | Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
database-1  | 2026-07-28 05:11:21+00:00 [Note] [Entrypoint]: Creating database mydb
database-1  |
database-1  | 2026-07-28 05:11:21+00:00 [Note] [Entrypoint]: Stopping temporary server
database-1  | 2026-07-28T05:11:21.692180Z 11 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.0.46).
database-1  | 2026-07-28T05:11:22.933899Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.46)  MySQL Community Server - GPL.
database-1  | 2026-07-28 05:11:23+00:00 [Note] [Entrypoint]: Temporary server stopped
database-1  |
database-1  | 2026-07-28 05:11:23+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
database-1  |
database-1  | 2026-07-28T05:11:23.896826Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1  | 2026-07-28T05:11:23.897127Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
database-1  | 2026-07-28T05:11:23.913685Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1  | 2026-07-28T05:11:24.037753Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1  | 2026-07-28T05:11:24.182254Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1  | 2026-07-28T05:11:24.182296Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1  | 2026-07-28T05:11:24.184182Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a diffe
rent directory.
database-1  | 2026-07-28T05:11:24.194654Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
database-1  | 2026-07-28T05:11:24.194727Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server -
 GPL.
```

```
C:\Users\15L\myapp>docker compose logs redis
redis-1  | Starting Redis Server
redis-1  | 1:C 28 Jul 2026 05:11:15.944 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
redis-1  | 1:C 28 Jul 2026 05:11:15.944 * Redis version=8.8.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1  | 1:C 28 Jul 2026 05:11:15.944 * Configuration loaded
redis-1  | 1:M 28 Jul 2026 05:11:15.944 * monotonic clock: POSIX clock_gettime
redis-1  | 1:M 28 Jul 2026 05:11:15.945 * Running mode=standalone, port=6379.
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf> RedisBloom version 8.8.2 (Git=unknown)
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf> Registering configuration options: [
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { bf-error-rate       :       0.01 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { bf-initial-size     :        100 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { bf-expansion-factor :          2 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { cf-bucket-size      :          2 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { cf-initial-size     :       1024 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { cf-max-iterations   :         20 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { cf-expansion-factor :          1 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf>   { cf-max-expansions   :         32 }
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * <bf> ]
redis-1  | 1:M 28 Jul 2026 05:11:15.947 * Module 'bf' loaded from /usr/local/lib/redis/modules//redisbloom.so
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> search-workers default: 16 (min of MAX_WORKER_THREADS=16 and CPU cores)
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> Redis version found by RedisSearch : 8.8.1 - oss
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> RediSearch version 8.8.0 (Git=d2ae026)
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> Low level api version 1 initialized successfully
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> gc: ON, prefix min length: 2, min word length to stem: 4, prefix max expansions: 200, query timeout (ms): 500, timeout policy
sor read size: 1000, cursor max idle (ms): 300000, max doctable size: 1000000, max number of search results:  1000000, default scorer: BM25STD,
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> Initialized thread pools!
redis-1  | 1:M 28 Jul 2026 05:11:15.956 * <search> Enabled workers threadpool of size 16
redis-1  | 1:M 28 Jul 2026 05:11:15.964 * <search> Subscribe to config changes
redis-1  | 1:M 28 Jul 2026 05:11:15.964 * <search> Subscribe to cluster slot migration events
redis-1  | 1:M 28 Jul 2026 05:11:15.964 * <search> Enabled role change notification
redis-1  | 1:M 28 Jul 2026 05:11:15.965 * <search> Cluster configuration: AUTO partitions, type: 0, coordinator timeout: 0ms
redis-1  | 1:M 28 Jul 2026 05:11:15.965 * Module 'search' loaded from /usr/local/lib/redis/modules//redisearch.so
redis-1  | 1:M 28 Jul 2026 05:11:15.967 * <timeseries> RedisTimeSeries version 80800, git_sha=42ca4f1078fca732b7f9256adbf25914d67e1cc9
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> Redis version found by RedisTimeSeries : 8.8.1 - oss
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> Registering configuration options: [
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-compaction-policy   :                }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-num-threads       :              3 }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-libmr-protocol    :       INTERNAL }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-retention-policy  :              0 }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-duplicate-policy  :          block }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-chunk-size-bytes  :           4096 }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-encoding          :     compressed }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-ignore-max-time-diff:           0 }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries>  { ts-ignore-max-val-diff :    0.000000 }
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> ]
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> Detected redis oss
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> Subscribe to ASM events
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * <timeseries> Enabled diskless replication
redis-1  | 1:M 28 Jul 2026 05:11:15.971 * Module 'timeseries' loaded from /usr/local/lib/redis/modules//redistimeseries.so
redis-1  | 1:M 28 Jul 2026 05:11:15.974 * <ReJSON> Created new data type 'ReJSON-RL'
```

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Frontend Service

```
C:\Users\15L\myapp>docker compose images
CONTAINER          REPOSITORY         TAG               PLATFORM          IMAGE ID        SIZE              CREATED
myapp-backend-1    myapp-backend      latest            linux/amd64       b1b45da00771    417MB             32 minutes ago
myapp-database-1   mysql              8.0               linux/amd64       7dcddc01f13b    234MB             2 months ago
myapp-frontend-1   myapp-frontend     latest            linux/amd64       01c9553d6eaa    63.1MB            32 minutes ago
myapp-redis-1      redis              latest            linux/amd64       c88d347edef6    54.3MB            3 days ago

C:\Users\15L\myapp>
```

```
C:\Users\15L\myapp>docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED          STATUS          PORTS                                                NAMES
1359f03286fa   nginx           "/docker-entrypoint.…"   6 minutes ago    Up 6 minutes    0.0.0.0:8081->80/tcp, [::]:8081->80/tcp              webserver
206bfc7fc073   mysql:8.0       "docker-entrypoint.s…"   6 minutes ago    Up 5 minutes    3306/tcp                                             mysqlserver
a9b760c62f74   myapp-frontend  "/docker-entrypoint.…"   21 minutes ago   Up 21 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp                  myapp-frontend-1
731574b1935a   myapp-backend   "python app.py"          21 minutes ago   Up 21 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp          myapp-backend-1
e9c6b430a3d2   redis:latest    "docker-entrypoint.s…"   21 minutes ago   Up 21 minutes   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp          myapp-redis-1
bc94eeb6adce   mysql:8.0       "docker-entrypoint.s…"   21 minutes ago   Up 21 minutes   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp          myapp-database-1
```

```
C:\Users\15L\myapp>docker compose ps
NAME               IMAGE              COMMAND                  SERVICE    CREATED          STATUS          PORTS
myapp-backend-1    myapp-backend      "python app.py"          backend    18 minutes ago   Up 18 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
myapp-database-1   mysql:8.0          "docker-entrypoint.s…"   database   18 minutes ago   Up 18 minutes   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
myapp-frontend-1   myapp-frontend     "/docker-entrypoint.…"   frontend   18 minutes ago   Up 18 minutes   0.0.0.0:80->80/tcp, [::]:80->80/tcp
myapp-redis-1      redis:latest       "docker-entrypoint.s…"   redis      18 minutes ago   Up 18 minutes   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
```

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Conclusion: Through this experiment, I learned how to deploy a multi-service application using Docker Compose by managing multiple containers through a single YAML configuration file. I understood how individual services such as the frontend, backend, database, and Redis cache work together while running in separate containers. This experiment also helped me learn how to build, start, monitor, and manage interconnected containers using Docker Compose commands, making the deployment process more organized, efficient, and easier to maintain.